

partial solns.

Name

- Show appropriate work.
 - Appropriate work for a linear ODE includes the integrating factor and some intermediate steps. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
 - Appropriate work for a linear system $y' = Ay + b$ includes a fully defined real complementary solution, appropriate undetermined coefficient forms, and some work indicating how needed linear systems are being constructed. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
 - Appropriate work for a higher order constant coefficient linear ODE includes the Characteristic Polynomial, appropriate undetermined coefficient forms, and some work indicating how linear systems are being constructed. Write down any linear systems you solve. Do not solve if asked for an Augmented Matrix.
- Explicitly real solutions are required.
- Computations
 - You are given Mathematica output for the the eigenvalues and eigenvectors A when asked to solve $y' = Ay + b$. Do not attempt to recompute eigenvalues etc.
 - You are given a factored form of the characteristic polynomial for polynomials of degree higher than 2. You need to compute roots of quadratics.
 - You know several ways to build an Augmented Matrix (AM) out of vectors and smaller matrices.
 - The matrix for the linear system $\sum_{i=1}^n c_i v_i = b_i$ is $A \cdot c = b$ where the columns of $A = (v_1 \dots v_n)$ are the vectors $v_1 v_2, \dots$ and an AM is $(A \mid b)$.
 - An AM for the linear system $\begin{matrix} A_{11} c_1 + A_{12} c_2 = b_1 \\ A_{21} c_1 + A_{22} c_2 = b_2 \end{matrix}$ is $\left(\begin{array}{cc|c} A_{11} & A_{12} & b_1 \\ A_{21} & A_{22} & b_2 \end{array} \right)$.
 - I_n is the $n \times n$ identity matrix with 1s on the diagonal and zeroes everywhere else.
 - Labelling and using eigenvalues/eigenvectors, roots etc. appropriately is a good idea.
- The exam has 5 ten point questions on four pages.
 - Q1 First order linear with I.C.
 - Q2 First Order Systems compute $y = y_c + y_p$ for an ODE system and AM to solve for given IC.
 - Q3 Higher Order ODEs. Like a chapter 4 HW question
 - Q4 Higher Order ODEs. Like a chapter 4 HW question
 - Q5 Higher Order ODEs. Like a chapter 4 HW question

Linear

#4

$$y' - \frac{\sin(t)}{\cos(t)} y = \frac{1}{\cos(t)}$$

$$\cos(t) y' - \sin(t) y = 1 \quad y(0) = 2$$

$$v = e^{-\int \frac{\sin(t)}{\cos(t)} dt}$$

$$u = \cos(t)$$

$$du = -\sin(t) dt$$

$$v = e^{\int \frac{1}{u} du} = e^{\ln(u)} = u = \cos(t)$$

$$(\cos(t) y)' = 1$$

$$\cos(t) y = t + C$$

$$\boxed{y(0) = 2}$$

$$\cos(0)(2) = 0 + C$$

$$\boxed{C = 2}$$

$$\boxed{y = \frac{t}{\cos(t)} + \frac{2}{\cos(t)}}$$

Linear ODEs $y' - 2y = 6e^{5t}$

1. Solve the ODE
- $2y'(t) - 4y(t) = 12e^{5t}$
- with
- $y(0) = 3$

$$e^{-2t} y' - 2e^{-2t} y = 6e^{3t} \Rightarrow e^{-2t} y = 2e^{3t} + C$$

$$y = 2e^{5t} + e^{2t}$$

2. Solve the ODE
- $t y'(t) - 4y(t) = 12t^5$
- with
- $y(1) = 3$
- .

$$y' - \frac{4}{t}y = 12t^4$$

$$v(t) = e^{-4 \ln(t)} = t^{-4}$$

$$t^{-4} y' - 4t^{-5} y = 12$$

$$y(1) = 3 \Rightarrow 1^{-4} 3 = 12 + C$$

$$(t^{-4} y)' = 12 \Rightarrow t^{-4} y = 12t + C$$

$$\Rightarrow y = 12t/e^4 - 9t^4$$

4. Solve the ODE
- $\cos(t) y'(t) - \sin(t) y(t) = 1$
- with
- $y(0) = 2$
- .

on sheet!

5. Solve the ODE
- $y'(t) + 2t y(t) = 6t e^{-t^2}$
- with
- $y(0) = 24$
- .

$$e^{t^2} y' + 2t e^{t^2} y = 6t \Rightarrow e^{t^2} y = 3t^2 + C$$

$$v(t) = e^{t^2}$$

$$e^0 24 = 0 + C$$

All with a box saying

$$y(t) = \frac{3t^2}{e^{t^2}} + \frac{24}{e^{t^2}}$$

Solving $y' = Ay + b$ with $y(0) = y_0$

1. Two parts

- 1.1. Write down an AM you would row reduce for the ODE
- $y' = Ay + b$
- with
- $y(0) = \begin{pmatrix} 12 \\ 11 \\ 10 \end{pmatrix}$

$$y_p = \begin{pmatrix} 3 + \cos(t) \\ 5 + \sin(t) \\ 6 + t \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} \cos(t) \\ 1 \\ 0 \end{pmatrix} - \sin(t) \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + c_3 e^{2t} \begin{pmatrix} \sin(t) \\ 1 \\ 0 \end{pmatrix} + \cos(t) \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$\text{AM} = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 12-4 \\ -2 & 1 & 0 & 11-5 \\ 1 & 0 & -2 & 10-6 \end{array} \right]$$

$$y(0) = y_c(0) + y_p(0)$$

$$\begin{pmatrix} 12 \\ 11 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} + c_1 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

- 1.2. Compute the solution of
- $y' = Ay + e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- where A is below

$$y_p = u e^{2t} \quad y_p' = 2u e^{2t}$$

$$2u e^{2t} = A u e^{2t} + e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$(2I - A)u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} -3 & -1 \\ -1 & -3 \end{pmatrix}$$

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -3 & 0 \end{bmatrix}$$

$$-3u_1 - u_2 = 1$$

$$-3(-3u_2) - u_2 = 1$$

$$u_2 = 1/8 \quad u_1 = -3/8$$

$$-u_1 - 3u_2 = 0$$

$$u_1 = -3u_2$$

$$y_p = \begin{pmatrix} -3/8 \\ 1/8 \end{pmatrix} e^{2t}$$

In[] := MatrixForm[Eigensystem[A = $\begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$]]

Out[] // MatrixForm =

$$\begin{pmatrix} 6 & 4 \\ \{1, 1\} & \{-1, 1\} \end{pmatrix}$$

2. Two parts

2.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 12 \\ 11 \end{pmatrix}$

$$y_p = \begin{pmatrix} 2 \sin(2t) \\ 5t + \cos(2t) \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$AM = \left[\begin{array}{cc|cc} 3 & 1 & 12 & 0 \\ 5 & 2 & 11 & -1 \end{array} \right]$$

$$y_c = c_1 e^{6t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$y = y_c + y_p = y_c + \frac{1}{8} e^{2t} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$y(0) = c_1 e^0 \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 e^0 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \sin(0) \\ 5(0) + \cos(0) \end{pmatrix}$$

$$c_1 \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 11 \end{pmatrix}$$

2.2. Compute the solution of $y' = Ay + e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ where A is below

$$a=5 \quad b=1$$

$$p = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, q = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

In[] := MatrixForm[Eigensystem[A = $\begin{pmatrix} 5 & -1 \\ 1 & 5 \end{pmatrix}$]]

Out[] // MatrixForm =

$$\begin{pmatrix} 5+i & 5-i \\ \{i, 1\} & \{-i, 1\} \end{pmatrix}$$

3. Two parts

3.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 12 \\ 11 \end{pmatrix}$

$$y_p = \begin{pmatrix} 2 \sin(2t) \\ 5t + \cos(2t) \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$AM = \left[\begin{array}{cc|cc} 3 & 1 & 12 & 0 \\ 5 & 2 & 11 & -1 \end{array} \right]$$

$$y(0) = y_c(0) + y_p(0)$$

$$\begin{pmatrix} 12 \\ 11 \end{pmatrix} = c_1 \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

3.2. Compute the complementary solution of

$$y' = Ay + e^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

write down the form of the particular solution and the AM you would solve to determine the coefficients where A is below.

$$a=5 \quad b=1 \quad p = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad q = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad v_3 = \begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix}$$

In[]:= MatrixForm[Eigensystem[A = $\begin{pmatrix} 5 & -1 & 1 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$]]

Out[]:= MatrixForm=

$\begin{pmatrix} 5+i & 5-i & 1 \\ (i, 1, 0) & (-i, 1, 0) & (-4, 1, 17) \end{pmatrix}$

$$y_c = c_1 e^{at} (\cos(bt)p - \sin(bt)q) + c_2 e^{at} (\sin(bt)p + \cos(bt)q) + c_3 e^t v_3$$

$$y_p = \frac{1}{10} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

$$AM = \left[\vec{p} \mid \vec{q} \mid \vec{v}_3 \mid y(0) - \frac{1}{10} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right]$$

$y_p = e^{2t} u \quad y_p' = 2e^{2t} u$

$2ue^{2t} = Aue^{2t} + e^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$[2I - A \mid \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}] \sim \left[\begin{array}{ccc|c} -3 & 1 & -1 & 1 \\ -1 & -3 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right]$

$\sim \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ -3 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right] \quad \begin{aligned} u_1 &= -3u_2 \\ 9u_2 + u_2 + 0 &= 1 \\ u_3 &= 0 \end{aligned}$

$u = \frac{1}{10} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$

$$y(0) = y_p(0) + y_c(0) = \frac{1}{10} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + c_1 \vec{p} + c_2 \vec{q} + c_3 \vec{v}_3$$

4. Two parts

4.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 12 \\ 11 \end{pmatrix}$

$$y_p = \begin{pmatrix} 2t + \cos(3t) \\ 7t - \sin(3t) \end{pmatrix} \text{ and } y_c = c_1 e^t \begin{pmatrix} 3 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$AM = \left(\begin{array}{c} \\ \\ \end{array} \right)$$

4.2. Compute the complementary solution of

$$y' = Ay + \sin(2t) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

write down the form of the particular solution and the AM you would solve to determine the coefficients where A is below.

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In[ ]:= MatrixForm[Eigensystem[A =  $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ ]]
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Out[ ]:= //MatrixForm=
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$$\begin{pmatrix} 3 & 1 & 1 \\ (1, 1, 0) & (-1, 1, 0) & (0, 0, 0) \end{pmatrix}$$

 $y_c =$
 $y_p =$
 $AM =$

Chapter 4 questions

There will be three 10 point questions on this material. Examples are below.

1. Two parts $\lambda^2 + 2\lambda - 3 = 0 \Rightarrow (\lambda+1)^2 - 4 = 0 \quad \lambda+1 = \pm 2 \quad \lambda = -3, 1$

1.1. Solve $y'' + 2y' - 3y = 10\sin(t)$

$$\begin{aligned} -4A + 2B &= 0 \\ -2A - 4B &= 10 \\ B &= 2A \\ -10A &= 10 \\ A &= -1, B = 2 \end{aligned}$$

$$y = C_1 e^{-3t} + C_2 e^t - \cos(t) - 2\sin(t)$$

$$\begin{aligned} y_p &= A \cos(t) + B \sin(t) \\ y_p' &= -A \sin(t) + B \cos(t) \\ y_p'' &= -A \cos(t) - B \sin(t) \\ -A \cos(t) - B \sin(t) + 2B \cos(t) - 3A \sin(t) &= 10 \sin(t) + 0 \cos(t) \\ -A \cos(t) + B \cos(t) - B \sin(t) - 3A \sin(t) &= 10 \sin(t) + 0 \cos(t) \end{aligned}$$

1.2. Write down the AM you would solve to determine the constants of integration for $y(0) = 1$ and $y'(0) = 3$.

$$\begin{aligned} y(0) = 1 &\Rightarrow C_1 + C_2 - \cos(0) = 1 \Rightarrow C_1 + C_2 = 2 \\ -3C_1 + C_2 + \sin(0) - 2\cos(0) &= 3 \Rightarrow -3C_1 + C_2 = 5 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ -3 & 1 & 1 & 5 \end{array} \right)$$

$$\text{AM} = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ -3 & 1 & 1 & 5 \end{array} \right]$$

2. Two parts

2.1. Solve $y'' - 2y' + 5y = 10e^{2t}$

$y =$

2.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$

$$\text{AM} = \left[\begin{array}{ccc|c} & & & \end{array} \right]$$

3. Two parts. Given

Expand $[(r+2)^2(r-1)]$

$r = -2$ twice, $r = 1$

3.1. Solve $y''' + 3y'' - 4y = 32e^{2t}$

$$y = \cancel{c_1 e^{-2t} + c_2 t e^{-2t}} + c_3 e^t + 2e^{2t}$$

$$\begin{aligned} y_p &= A e^{2t} \\ y_p' &= 2A e^{2t} \\ y_p'' &= 4A e^{2t} \\ y_p''' &= 8A e^{2t} \end{aligned}$$

$$8A + 3(4A) - 4A = 32$$

$$16A = 32$$

$$\boxed{A = 2}$$

3.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

$$\text{AM} = \left[\begin{array}{ccc|c} 1 & 0 & 1 & -4-2 \\ -2 & 1 & 1 & 5-4 \\ 4 & -4 & 1 & 3-8 \end{array} \right]$$

$$\begin{aligned} c_1 + c_2(0) + c_3 + 2 &= -4 \\ -2c_1 + c_2 + c_3 + 4 &= 5 \\ 4c_1 - 4c_2 + c_3 + 8 &= 3 \end{aligned}$$

4. Two parts. Given

$$\begin{aligned} y' &= -2c_1 e^{-2t} - 2c_2 t e^{-2t} + c_3 e^t + 4e^{2t} \\ y'' &= 4c_1 e^{-2t} + 4c_2 t e^{-2t} - 2c_2 e^{-2t} - 2c_2 e^{-2t} + c_3 e^t + 8e^{2t} \end{aligned}$$

In[]:= Expand[(r²+4)(r-1)]
 Out[]:= -4+4r-r²+r³

$r^2 = -4$ or $r = \pm 2i$ or $r = 1$

4.1. Solve $y''' - y'' + 4y' - 4y = 32e^{-t}$

$$y = c_1 \cos(2t) + c_2 \sin(2t) + c_3 e^t - 3.2e^{-t}$$

$$\begin{aligned} y_p &= Ae^{-t} \\ y_p' &= -Ae^{-t} \\ y_p'' &= Ae^{-t} \\ y_p''' &= -Ae^{-t} \end{aligned}$$

$$\begin{aligned} -A - A - 4A - 4A &= 32 \\ -10A &= 32 \quad A = -3.2 \end{aligned}$$

4.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

$$AM = \begin{bmatrix} 1 & 0 & 1 & 1 & -4+3.2 \\ 0 & 2 & 1 & 1 & 5-3.2 \\ -4 & 0 & 1 & 1 & 3+3.2 \end{bmatrix}$$

$$\begin{aligned} c_1 + 0c_2 + c_3 - 3.2 &= -4 \\ 0c_1 + 2c_2 + c_3 + 3.2 &= 5 \\ -4c_1 + 0c_2 + c_3 - 3.2 &= 3 \end{aligned}$$

$$\begin{aligned} y' &= -2c_1 \sin(2t) + 2c_2 \cos(2t) + c_3 e^t + 3.2e^{-t} \\ y'' &= -4c_1 \cos(2t) - 4c_2 \sin(2t) + c_3 e^t - 3.2e^{-t} \end{aligned}$$

5. Two parts. Given

In[]:= Expand[(r+1)³]
 Out[]:= 1+3r+3r²+r³

$r = -1, -1, -1$

5.1. Solve $y''' + 3y'' + 3y' + y = 8e^t$

$$y = c_1 e^{-t} + c_2 t e^{-t} + c_3 t^2 e^{-t} + e^t$$

$$\begin{aligned} y_p &= Ae^t \\ y_p' &= Ae^t \\ y_p'' &= Ae^t \\ y_p''' &= Ae^t \end{aligned}$$

$$A + 3A + 3A + A = 8 \quad A = 1$$

5.2. Write down the AM you would solve to determine the constants of integration for $y(0) = -4$ and $y'(0) = 5$ and $y''(0) = 3$

$$AM = \begin{bmatrix} 1 & 0 & 0 & 1 & -4-1 \\ -1 & 1 & 0 & 1 & 5-1 \\ 1 & -2 & 2 & 1 & 3-1 \end{bmatrix}$$

$$\begin{aligned} c_1 + 0c_2 + 0c_3 + 1 &= -4 \\ -c_1 + c_2 + 0c_3 + 1 &= 5 \\ c_1 - 2c_2 + 2c_3 + 1 &= 3 \end{aligned}$$

6. Two parts. Given

In[]:= Expand[(r-2)(r+1)(r+3)]
 Out[]:= -6-5r+2r²+r³

$$\begin{aligned} y' &= -c_1 e^{-t} + c_2 e^{-t} - c_2 t e^{-t} + c_3 2t e^{-t} - c_3 t^2 e^{-t} + e^t \\ y'' &= c_1 e^{-t} - c_2 e^{-t} - c_2 e^{-t} + c_2 t e^{-t} + c_3 2e^{-t} - c_3 2t e^{-t} + c_3 t^2 e^{-t} - c_3 2t e^{-t} \end{aligned}$$

$$r = 2, -1, -3$$

Solve $y''' + 2y'' - 5y' - 6y = 3t$

$$y = c_1 e^{2t} + c_2 e^{-t} + c_3 e^{-3t} - \frac{t}{2} + \frac{5}{12}$$

guess $y_p = at + b$

$$y_p' = a + 0$$

$$y_p'' = 0, y_p''' = 0$$

$$-5a - 6(at + b) = 3t + 0$$

$$-6a = 3$$

$$a = -\frac{1}{2}$$

$$-5a - 6b = 0 \quad 6b = -5a = \frac{5}{2}$$

$$b = \frac{5}{12}$$

6.2. Write down the AM you would solve to determine the constants of integration for $y(0) = 2$ and $y'(0) = -5$ and $y''(0) = 1$

$$\text{AM} = \begin{pmatrix} 1 & 1 & 1 & 2 - \frac{5}{12} \\ 2 & -1 & -3 & -5 + \frac{1}{2} \\ 4 & 1 & 9 & 1 \end{pmatrix}$$

$$\begin{aligned} c_1 + c_2 + c_3 - \frac{0}{2} + \frac{5}{12} &= 2 \\ 2c_1 - c_2 - 3c_3 - \frac{1}{2} &= -5 \\ 4c_1 + c_2 + 9c_3 + 0 &= 1 \end{aligned}$$

$$\begin{aligned} y' &= 2c_1 e^{2t} - c_2 e^{-t} - 3c_3 e^{-3t} - \frac{1}{2} \\ y'' &= 4c_1 e^{2t} + c_2 e^{-t} + 9c_3 e^{-3t} + 0 \end{aligned}$$